

Product Video Demo & Walkthrough Script

Target Duration: 3 Minutes 30 Seconds | **Format:** Screen Recording + Narration Voiceover | **Target Audience:** Hackathon Judges (Devpost), Educators, Students

Core Narrative: Students face a crisis of fragmented learning—shuffling between dense textbook PDFs and endless YouTube video lectures, while generic AI chats hallucinate formulas without syllabus grounding. **Gyaan Setu** solves this by ingesting **both documents and YouTube lectures** into verifiable, timestamped knowledge units, extracting a curriculum **Knowledge DAG**, and guiding students through a **strictly grounded Socratic Tutor**, **Bayesian Knowledge Tracing (BKT)** adaptive quizzes, and **Spaced Repetition (SM-2)** audio briefs.

Recommended Demo Preparation Checklist

Step	File / Resource to Prepare	Expected Demo Action
1. Course PDF	Any syllabus PDF (e.g. <i>Machine Learning Guide.pdf</i>)	Drag & drop into Library → Upload Course Files
2. YouTube URL	E.g. https://www.youtube.com/watch?v=aircArvnKk (Neural Networks by 3Blue1Brown)	Paste into Library → Ingest YouTube / Video Lecture
3. Clean State	Database already reset to clean slate (verified 0 rows)	Ready for live upload during the video demo
4. Servers	Backend on :8000, Frontend on :3000	Both running and verified healthy

Scene-by-Scene Production Script

Follow each scene sequentially. The timing is calibrated for a natural speaking rate of 130-140 words per minute.

SCENE 1 00:00 – 00:22 (22s)	The Hook & The Crisis in Modern Learning Visual Action: Open on Gyaan Setu Landing Page / Dashboard (http://localhost:3000). Show clean Greek-key architectural accents, gold cards, and the 5 navigation tabs: Library, Tutor, Practice, Revise, Progress.	Spoken Narration (Voiceover): <i>"Every student today faces the same struggle: our study materials are completely fragmented. Textbook PDFs, slide decks, and hour-long YouTube lecture videos live in silos. And when we ask generic AI chatbots for help, they hallucinate formulas and ignore our actual syllabus.</i> <i>Meet Gyaan Setu—the grounded multimodal AI tutor and curriculum engine that unifies your documents and video lectures into one verifiable learning system."</i>
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SCENE 2 00:22 – 00:45 (23s)	Dual Ingestion Part 1: Ingesting Course Documents (PDF) Visual Action: Click Library tab. In the Upload Course Files section, drag and drop a course PDF (e.g. <i>Machine Learning Guide.pdf</i>). Show the real-time progress bar analyzing layout and indexing units into PyMuPDF chunks. Point out the newly created source card showing Page Count and unit count.	Spoken Narration (Voiceover): <i>"Let's start in the Library. First, we upload our course textbook PDF. Gyaan Setu's multimodal ingestion pipeline extracts every page, table, and formula, indexing them into traceable units with exact page numbers. No lost context, no unverified chunks."</i>
SCENE 3 00:45 – 01:10 (25s)	Dual Ingestion Part 2: Ingesting YouTube Lecture Videos Visual Action: Switch to the Ingest YouTube / Video Lecture tab. Paste a YouTube link (e.g. 3Blue1Brown Neural Networks). Enter title 'Neural Networks Foundation'. Click Ingest Video Lecture . Show notification: 'Successfully ingested video units'. Notice the exact timestamped units (■■ MM:SS–MM:SS) in the source viewer.	Spoken Narration (Voiceover): <i>"Now for the magic: YouTube video lectures. Simply switch to the YouTube tab and paste any lecture URL. Gyaan Setu extracts timestamped transcripts—down to the exact second. Whether a concept was taught on page 4 of the textbook or at minute 3:45 of a video lecture, both are now indexed into our unified semantic vector space."</i>
SCENE 4 01:10 – 01:30 (20s)	Building the Curriculum Knowledge DAG Visual Action: Click the gold button: Build Knowledge DAG . Show the toast: 'Knowledge DAG built! Topics indexed'. Click Progress tab to show the interactive Directed Acyclic Graph (DAG) with canvas pan/zoom, prerequisite arrows, and topological learning sequence.	Spoken Narration (Voiceover): <i>"Next, with one click, we build the Curriculum Knowledge DAG. Using NetworkX, Gyaan Setu maps prerequisite dependencies between concepts, detecting cycles and organizing a topological learning roadmap. Students never get stuck on an advanced topic without mastering the prerequisite fundamentals."</i>
SCENE 5 01:30 – 02:05 (35s)	Source-Grounded Socratic Tutor & Dual Citations Jump Visual Action: Navigate to Tutor tab. Toggle Strict Grounding on. Type a Hinglish/English question: 'What is the difference between Gradient Descent and Backpropagation?' Press Send. Show the grounded answer streaming back with citation badges: [Page 4] and [03:45]. CRITICAL INTERACTIVE ACTION: 1. Click [Page 4] → Modal opens showing textbook excerpt! 2. Click [03:45] → Modal opens showing embedded YouTube player playing at that exact second!	Spoken Narration (Voiceover): <i>"Now, let's learn with our Socratic AI Tutor. We ask: 'What is the difference between Gradient Descent and Backpropagation?'"</i> <i>Notice how every single claim is strictly verified with clickable citation tags. When I click [Page 4], the source textbook modal opens immediately with the exact excerpt. And when I click [03:45], it embeds the YouTube lecture player and jumps directly to that second! Zero hallucinations. Total academic trust."</i>

SCENE 6 02:05 – 02:25 (20s)	Multilingual Voice & Speech Tutoring Visual Action: In Tutor, click the Voice Chat / Microphone button. Speak a question in Hindi or English (e.g. 'Mujhe loss function ka intuitive matlab batao'). Show speech transcription via faster-whisper and hear the instant Edge-TTS audio response play with audio waveform indicator.	Spoken Narration (Voiceover): <i>"Gyaan Setu also features real-time voice tutoring. Powered by faster-whisper STT and Edge-TTS synthesis, students can speak naturally in English, Hindi, or Hinglish. Vernacular queries are expanded into academic terminology for seamless cross-lingual retrieval."</i>
SCENE 7 02:25 – 02:50 (25s)	Adaptive Practice Quiz & Bayesian Knowledge Tracing (BKT) Visual Action: Click Practice tab. Select topic or click Start Adaptive Quiz . Show an MCQ generated from the ingested PDF and video lecture. Select an option and submit. Show immediate verification feedback, source citations, and the Mastery Probability P(L) update powered by BKT.	Spoken Narration (Voiceover): <i>"How do we test our understanding? The Practice module generates grounded MCQs and short-answer questions verified against our units. When we submit an answer, our cognitive Bayesian Knowledge Tracing engine updates our latent knowledge probability P(L) and memory stability in real time—adapting future quizzes to strengthen our weak spots."</i>
SCENE 8 02:50 – 03:15 (25s)	Spaced Repetition (SM-2) Flashcards & 2-Minute Audio Brief Visual Action: Click Revise tab. Show the SuperMemo SM-2 flashcard. Click to flip card and rate memory recall (0 to 5). Then scroll down to the Audio Synthesis Studio , click Play 2-Min Audio Brief or Podcast Mode to hear the generated two-host study podcast.	Spoken Narration (Voiceover): <i>"For retention, the Revise module schedules flashcards using the SuperMemo SM-2 spaced repetition algorithm, calculating optimal review intervals. And when you're on the commute, listen to an automated 2-minute Audio Brief or a two-host dialogue podcast generated directly from your syllabus!"</i>
SCENE 9 03:15 – 03:35 (20s)	Architecture, Zero Cost Ledger & Closing Visual Action: Return to Dashboard / Progress view. Show the Mastery status and highlight the live backend metrics: 45 API endpoints, local ChromaDB, local sentence-transformers, and zero external API costs incurred in the usage ledger.	Spoken Narration (Voiceover): <i>"Gyaan Setu is built with production-grade engineering: FastAPI, SQLite in WAL mode, local ChromaDB embeddings, and a Next.js 16 frontend. Best of all, it runs with zero external API costs, providing accessible, private, and syllabus-grounded education to every student.</i> <i>Thank you for watching Gyaan Setu—bridging curiosity to true mastery."</i>

■ Video Recording & Production Best Practices

- **Screen Resolution & Aspect Ratio:** Record at standard **1080p (1920x1080) at 60 FPS**. Set browser zoom to **100% or 110%** so gold UI cards, buttons, and Greek-key borders are crisp and legible on mobile screens.

- **Microphone & Audio Delivery:** Use a USB condenser mic or headset. Speak with an **enthusiastic, confident, and articulate** pacing (**130-140 words per minute**). Leave 1-2 seconds of silence before and after each scene for clean audio splicing.

- **Cursor Movement & Highlights:** Move the mouse smoothly with purpose. Avoid rapid circle-spinning. Hover over citations ([Page 4], [03:45]) for a full second before clicking to draw the viewer's eye.

- **Interactive Citation Climax:** In Scene 5, ensure the audience clearly sees the YouTube video modal pop up and start playback at the specified timestamp. This is the **'WOW' moment** for judges proving true multimodal grounding.

- **B-Roll / Title Cards (Optional):** Add a 2-second intro title card with the Gyaan Setu logo and tagline: *'Grounded Multimodal AI Tutor & Cognitive Curriculum Engine'*, and a 3-second closing card with your GitHub repository link.

■ Key Hackathon Judging Points to Emphasize

- ✓ **Verifiable Grounding vs. Generic RAG:** Gyaan Setu doesn't just return chunk snippets; it verifies chunks across PyMuPDF and YouTube transcripts, appending enforceable page and second-level citations.

- ✓ **Cognitive Science Foundation:** Features true Bayesian Knowledge Tracing (BKT) updating latent knowledge probability $P(L)$ and SuperMemo SM-2 spaced repetition stability rather than superficial scoring.

- ✓ **Curriculum DAG Reasoning:** Uses NetworkX Directed Acyclic Graphs with cycle detection and topological sorting to prevent out-of-order learning.

- ✓ **Zero External API Cost Architecture:** Demonstrates that high-end AI education can be delivered affordably using local sentence-transformers, ChromaDB, Edge-TTS, and faster-whisper.